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MIL-T-45542D
1 JULY 1998
SUPERSEDING
MIL-T-45542C
1 JUNE 1984

MILITARY SPECIFICATION

TOOL SETS, SHOP SETS AND KITS

MAINTENANCE, MODIFICATION AND TOOL

PACKAGING OF

This specification is approved by use for all Department and Agencies of the Department of Defence.

1 SCOPE

1.1 Scope. This specification covers the requirements for methods of preservation, packing, and marking of components for both common and special tool sets, shop sets, and kits (hardware, installation, maintenance, modification, and tool) for handling, shipment, and storage. (This specification does not include tool sets or tool kits that are furnished in plastic, cloth, or leather cases, or relatively small wooden or metal carrying cases) (see 6.4).

1.2 Classification. Tool sets, kits, and shop sets shall be of the following types, as specified (see 6.2.1).

Type I - Kits and sets shall be the complete assemblage of tools, packaged in a single carrying case.

Type II - Kits and sets shall be the complete assemblage of tools, equipment, and compressed gases requiring two or more shipping containers.

Type III-Kits and sets shall be the complete assemblage of tools and equipment requiring a single multi-packed box (less carrying case).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: SMCAR-EST-A, Rock Island, IL 61299 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defence Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.1).

SPECIFICATIONS

FEDERAL

A-A-883	- Tape, Pressure Sensitive Adhesive, Masking
UU-T-81	- Tags, Shipping and Stock
PPP-B-566	- Boxes, Folding, Paperboard
PPP-B-585	- Boxes, Wood, Wirebound
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-wall
PPP-B-676	- Boxes, Setup
PPP-B-1055	- Barrier Material, Waterproofed, Flexible
PPP-C-795	- Cushioning Material, Flexible, Cellular, Plastic film for Packaging Applications
PPP-C-843	- Cushioning Material, Cellulosic
PPP-C-1120	- Cushioning Material, Uncompressed Bound Fiber for Packaging
PPP-C-1797	- Cushioning Material, Resilient, Low Density, Uncellular, Polypropylene Foam
PPP-H-1581	- Hardware (Fasteners and Related Items) Packaging and Packing for Shipment and Storage of
PPP-P-40	- Preservation and packaging of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery
PPP-P-115	- Pads, Paper Macerated
PPP-P-291	- Paperboard, Wrapping and Cushioning
PPP-T-60	- Tape, Packaging Waterproof

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MILITARY

MIL-P-116 - Preservation, Methods of
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-F-3296 - Forges, Furnaces and Ovens (Exclusive of Space Heating and Cooking), Packaging of
MIL-P-3420 - Packaging Materials, Volitile Corrosion Inhibitor Treated, Opaque
MIL-C-3600 - Compressors, Rotary Power-Driven; and Compressors Receiprocating, Power-Driven; Air and Gas (except Oxygen and Refrigerant
MIL-C-3774 - Crates, Wood; Open 12,000- and 16,000-Pound Capacity
MIL-B-3865 - Blocks, Rope, Tackle: Packaging of
MIL-C-5501 - Cap and Plug, Protective, Dust and Moisture Seal General Specification for
MIL-I-8574 - Inhibitors, Corrosion, Volatile, Utilization of
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of
MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of
MIL-G-28554 - Generator Sets, Mobil Electric Power and Supplemental Equipment, Packaging of
MIL-B-43666 - Boxes, Shipping Consolidation
MIL-V-45554 - Vulcanizing Equipment, Including Related Items, Tire and Tube, Rebuild and Repair; Packaging of
MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of
MIL-B-45977 - Bins, Cabinets, Shelving, and Worktables, Packaging of
MIL-C-52950 - Crates, Wood, Open and Covered

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing: With Appropriate Test Methods
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking
MIL-STD-2073-1- DOD Material, Procedures For Development and Application of Packaging Requirements

'(Unless otherwise indicated, copies of federal and military specifications and standards, are available from Standardization Documents Order Desk, Building 4D, 700 Robins Avenue, Philadelphia PA 19111-5094)'.

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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

TM 38-250/AFR 71-4/NAVSUPPUB 505/MCOP 4030.19E/DLAM 4145.3 - Preparation of Hazardous Materials for Military Air Shipment

DOD 4145.19-R-1 - Storage and Materials Handling

DOD 4500.32-R - Military Standard Transportation and Movement Procedures

(Copies of specification, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2.1).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951-82 -Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for testing Materials, 1916 Race Street, Philadelphia, PA 19103

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

TARIFF BOE-6000-I - Bureau of Explosives Tariffs (CFR 49, Parts 106-177)

(Application for copies should be addressed to Thomas A. Phemister Agent, Association of American Railroads, Bureau of Explosives, 1920 'L' Street N.W., Washington, D.C. 20036).

INTERNATIONAL CIVAL AVIATION ORGANIZATION (ICAO)

(Application for copies should be addressed to International Regulations Publishing and Distributing Organization, P.O. Box 66165, Chicago IL 60666)

INTERNATIONAL MARITIME ORGANIZATION (IMO)

(Application for copies should be addressed to International Maritime Organization, 101-104 Picadilly London WIV OAE, England

INTERNATIONAL AIR TRANSPORT ASSOCIATION (IATA)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the supplier shall furnish one complete pack for first article inspection. Whenever first article inspection is specified or performed, unit pack examination and tests for packaging may be performed also. The first article may be either a preproduction pack or an initial production pack which conforms to the requirements of this specification. In either case, the approved first article and the production packs shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment and packaging in accordance with the requirements of this specification.

3.2 Component grouping. For the purpose of this specification, component parts and assemblies have been segregated into groups according to the composite characteristics as specified in Table I in order to determine the applicable method protection required.

Table I. Composite characteristics component grouping

Group	Composition	Description
1	Metallic, Ferrous	Items in this group are comprised wholly of metals and alloys containing an appreciable quantity of iron, including the many types of steel. Surfaces may be bare, partially plated, coated, or painted. Items of this nature include, but are not limited to , hoist, tow chains, bolt cutters, workbenches, anvils, tow bars, auto jacks, vises, and common hand tools.
2	Metallic, including Non-ferrous and Painted or Plated Ferrous	Items in this group are comprised wholly of metallic, nonferrous metals and items which are completely painted or plated of this nature include, but are not limited to, nonsparking hand tools such as aluminum tipped hammers. Also, folding tables, crowbars, and tool cabinets.

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Table I. Composite Characteristics Components Grouping-continued

3	Metallic, and Nonmetallic Composites	Items in this group are comprised of two or more materials, one of which being metallic and the remaining materials nonmetallic. Items of this nature include, but are not limited to, tire gages, electric soldering irons, files and reamers with wooden handles, electric lanterns, hand saws, claw hammers, axes, and wire brushes.
4	Nonmetallic	Items in this group are comprised of nonmetallic materials regardless of whether the item is comprised wholly of one nonmetallic material or a combination of nonmetallic materials. Items of this nature include, but are not limited to, technical data, leather and rubber aprons, tools bags, goggles, safety gloves, and abrasive products.
5	Electrical and Electronic Equipment	Items in this group include electronic equipment. Items of this nature include, but are not limited to, test sets, electric hand tools, motors, and complete electronic systems.
6	Pneumatic Hand Tools	Items in this group are tools that are air power driven and include , but are not limited to, hammers, sanders, polishers, buffing tools, riveters, and impact wrenches.
7	Complex Items	Items included in this group are individually preserved, packaged, and packed because of their construction, configuration, size, and weight. Items of this nature include, but are not limited to, metal and woodworking machines, compressors, cleaners, and sewing machines.
8	Compressed Gas containers	Items in this group are limited to compressed gas in containers.

3.3 Materials. Materials used in preservation, packaging, and packing of tool sets, shop sets and kits shall be as specified herein. Materials shall be new and free from defects and imperfections which may affect the serviceability of the complete package or pack.

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3.4 Disassembly. When practical, for the protection of components, attachments and accessories against damage, pilferage or to reduce cubage, items may be disassembled only to a necessary basic degree. Disassembled parts shall be clearly and legibly marked as to identity and proper location on the assembled item. All fasteners removed during disassembly shall be secured in one of the mating parts. A part shall not be removed from an assembly unless it can be reassembled readily in the field without special skills or tools.

3.5 Disassembled parts marking. Disassembled parts shall be marked, when necessary with instructions to facilitate reassembly. Removed parts and mating parts on the item shall be marked by stenciling on the part, or by use of tags. Tags shall be waterproofed and ink used (on parts or tags) shall be waterproof. Tags shall be firmly secured to each individual part.

3.6 Processing conditions. Preservation shall be accomplished within buildings which have facilities to limit the accumulation of dust and moisture on the equipment being processed. Equipment brought into processing areas from the outside shall be allowed sufficient time to reach ambient temperatures before processing. Cleaning, drying, application of preservatives, and sealing of barriers shall progress in an uninterrupted series of operations. When periods of interruption are necessary temporary protection shall be provided to partially processed items, as required, to ensure against contamination.

3.6.1 Cleaning and drying. Thorough cleaning and drying shall be accomplished without damage to any part. Cleaning of all exposed surfaces of parts and assemblies shall be accomplished through one or a combination of processes referenced in MIL-P-116. Drying shall be accomplished through one or more drying procedures referenced therein. Special cleaning processes and drying procedures may be employed upon approval of the procuring activity concerned.

3.6.2 Preservatives. Preservatives specified herein, identified by symbol, e.g., type P-1, shall conform to the applicable specifications listed in MIL-P-116.

3.6.3 Stowage compartments. When stowage compartments are provided in the carrying case, the kit components shall be stowed within the designated compartments to the extent practicable. Overflow of packaged items shall be placed in tiers inside the carrying case except when the carrying case will not adequately house the complete packaged kit, the overflow of parts will be placed inside the unit container (see 3.7.1.1.5) along with the carrying case.

3.6.4 Unit protection. The methods of preservation specified herein shall be in accordance with MIL-P-116.

3.7 Preservation. Preservation shall be level A, C or Commercial as specified (see 6.2.1).

3.7.1 Level A. Components of tool sets and kits shall be preserved and unit packaged in accordance with methods specified herein. Tool and equipment items shall be unit packaged in quantities specified in the applicable packaging data document. Alternatively, diminutive (small) type items, bearing the same National Stock Number, may be unit packaged in multiple quantities with the following limitations:

a. Maximum size of 24 cubic inches for individual items.
b. Maximum of 100 pieces per unit package.
c. Maximum net load of 10 pounds.

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The unit method of protection specified in the packaging data document shall in no case be down graded when packaging items in multiple quantities. However, the submethod may be changed to better accommodate the load, e.g., submethod IC-1 to IC-2 and IA-8 to IA-15.

3.7.1.1 Type I. Type I kits or tools sets shall be preserved and packaged in accordance with the applicable methods listed below. The selection of methods and application of materials shall be in accordance with the criteria and guidelines of MIL-P-116 and MIL-STD-2073-1.

3.7.1.1.1 Method I utilizing Volative Corrosion Inhibitor (VCI) (P-18). The preferred method of packaging type I kits or tools sets is method I utilizing VCI treated materials, in place of a contact preservative. The VCI material shall conform to MIL-P-3420, and be used in accordance with 'Limitation on use' in MIL-I-8574. Kits or tool sets packaged by this method will be mainly comprised of items in groups 1,3 and 4 or a combination thereof. (This method shall not be used for items in group 5.

3.7.1.1.2 Method I utilizing contact preservative. Component parts that cannot be packaged with VCI treated materials, because of their composition or surface finish (see MIL-I-8574), shall be packaged in accordance with method I, utilizing a contact preservative. Component parts that are usable without removing the preservative shall be coated with type P-1 or P-19 preservative. Thoroughly dried parts need not be wrapped. Unless otherwise specified in commodity or packaging specifications, critical ferrous metal surfaces shall be preserved with type P-2 or type P-6 preservative. Surfaces preserved with type P-2 or type P-6 preservative shall be wrapped with barrier material conforming to MIL-B-121, grade A, class 2. Barrier material shall be secured with tape conforming to A-A-883. Tools preserved by this method will be mainly comprised of items in groups 1,3, and 6.

3.7.1.1.3 Method II. Unless otherwise specified in commodity or packaging specifications, kits or tool sets comprised of items in group 5 shall be packaged in accordance with the applicable submethod of method II. The carrying case may serve as the inner container when submethods IIb and IIe are employed (3.6.4).

3.7.1.1.4 Method III. Method III shall be utilized in packaging type I kits or tool sets comprised of items in group 2.

3.7.1.1.5 Unit container. Type I kits and sets shall be individually packaged in a snug-fitting fiberboard box conforming to PPP-B-636, class weather-resistant. The box closure shall be in accordance with the appendix to the box specification.

3.7.1.1.6 Cushioning. Cushioning material shall be employed inside the carrying case to restrict movement and protect parts that are vulnerable to damage. Material shall conform to any of the following: PPP-C-843, PPP-C-1797, PPP-C-795, PPP-C-1120, PPP-P-115, or PPP-P-291. All fragile and delicate items, or items having cutting edges, shall be isolated to prevent impact with other components.

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3.7.1.2 Type II and III. Components of types II and III tool sets and shop sets, categorized by composition characteristics of the various groups of Table I, shall be preserved and packaged as follows:

Group 1. Items having composition characteristics of this group shall be preserved and packaged in accordance with level A requirements of MIL-W-45562, MIL-B-3865, PPP-H-1581, or PPP-P-40, as applicable. In the event the specification does not cover the item. Package in accordance with method I of MIL-P-116.

Group 2. Items having composition characteristics of this group shall be preserved and packaged in accordance with level A requirements of PPP-H-1581, MIL-B-45977, MIL-W-45562, or PPP-P-40, as applicable. In the event the packaging specification does not cover the item, package in accordance with method III of MIL-P-116.

Group 3. Items having composition characteristics of this group shall be preserved and packaged in accordance with level A requirements of PPP-P-40, MIL-B-45977, MIL-B-3865, or MIL-W-45562, as applicable. In the event the packaging specification does not cover the item, package delicate and fragile items in accordance with method IC and rugged-type items method I of MIL-P-116.

Group 4. Items having composition characteristics of this group shall be preserved and packaged in accordance with level A requirements of PPP-P-40. In the event the packaging specification does not cover the item, package in accordance with IC of MIL-P-116.

Group 5. Items having composition characteristics of this group shall be preserved and packaged in accordance with level A requirements of MIL-E-17555, MIL-E-16298, PPP-P-40, or the commodity specification, as applicable. In the event the packaging specification does not cover the item, package in accordance with method II of MIL-P-116.

Group 6. Items having composition characteristics of this group shall be preserved and packaged in accordance with level A requirements of PPP-P-40.

Group 7. Items having composition characteristics of this group shall be preserved and packaged in accordance with level A requirements of MIL-M-18058, MIL-F-3296, MIL-C-3608, MIL-E-16298, MIL-G-28554, or the applicable commodity specification.

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Group 8. Items having composition characteristics of this group shall be packaged in accordance with Tariff BO E-6000-I (CFR 49, Parts 166-177) for geographical North America shipments; International Maritime Organization (IMO) for international vessel; International Air Transport Association (IATA) for international air; International Civil Aviation Organization (IACO), and AFR 71-4/TM 38-250/NAVSUPPUB 505/MCOP 4030.19E/ DLAM 4145.3 for Military air shipments.

Transportation through the Defence Transportation System is covered in DOD 4500.32-R. In addition to the above, packaging requirements for cylinders must have their valves protected by one of the following methods:

a. By a securely attached metal cap of sufficient strength to protect the valve from injury during transit.
b. By boxing or crating the cylinder or sphere so as to give proper protection to the valve.
c. By a recessed valve or otherwise by a protective valve so that it will be protected during transit.

3.7.1.3 Alternate method of preservation. Ferrous metal items which are received or stored at a DOD installation, that are coated with a preservative compound, other than that specified by the applicable packaging data document, or coated with lacquer or varnish, will be acceptable for level A protection, unless evidence of deterioration or corrosion is present.

3.7.1.4 Alternate method of unit protection. For set components which are scheduled to be unit packaged method I or method III, that utilize a fiberboard, wood, or paperboard box as the unit container, unit containers may be omitted, if the consolidation container is water-resistant (see 3.9.1.4), or when the item is packed as specified in 3.9.1.3. If unit container is not omitted the use of Domestic Grade Fiberboard is acceptable for level A protection when packed within Level A consolidation container (see 3.9.1.4). The water-resistant protection will come from the exterior consolidation container in lieu of the unit container. Critical or painted surfaces may require additional protection against abrasion or physical damage. These surfaces shall be protected by wrapping or covering with cushioning material specified in 3.7.1.1.6; or package the item in a multi-pack container, or cover the surface with tape or cap-plugs conforming to PPP-T-60, type optional or MIL-C-5501, respectively.

3.7.2 Level C preservation. Level C preservation of tool sets, shop sets, and kits shall conform to the requirements of MIL-STD-1188.

3.7.3 Commercial preservation. Commercial preservation of tool sets, shop sets, and kits shall conform to the requirements of ASTM D 3951-82.

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3.7.4 Army procurements.

3.7.4.1 Level A unit packaging. Unit packs shall be as specified except that all unit packs in fiberboard containers shall be class weather-resistant.

Level A	Remarks
IA-8	
IA-14	Omit outer container and wrap with Kraft paper.
II	Omit desiccant
III	No change.

3.8 Intermediate packaging. Type I kits or tool sets weighing ten pounds or less, bearing the same stock number, shall be intermediate packaged in a close-fitting box conforming to PPP-B-636, type CF, class weather-resistant. The gross weight of the intermediate package shall not exceed 35 pounds nor displace more than 1.5 cubic feet. Box closure shall be in accordance with the appendix to the box specification.

3.8.1 Level A intermediate packs. All intermediate containers shall either be weather (or water) resistant or overwrapped with waterproof barrier materials. Containers conforming to PPP-B-566 or PPP-B-676 shall be variety 2, or be overwrapped with waterproof barrier materials. Intermediate containers shall be uniform in size, shape, and quantities; shall be of minimum weight and cube; and shall contain multiples of 5 unit packs not to exceed 100 unit packs. Intermediate containers shall not exceed a maximum of 40 pounds net weight and a maximum of 1.5 cubic feet, with at least each of two dimensions not exceeding 16 inches. Intermediate containers shall not be required when the total quantity being shipped will result in only one intermediate pack per shipping container.

3.9 Packing. Packing shall be level A, B, C or Commercial specified (see 6.2.1).

3.9.1 Level A.

3.9.1.1 Type I. Kits or tool sets shall be packed in a box conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-585, class 3, style 2. Box closure and strapping requirements shall be in accordance with the applicable box specification and the appendix thereto. Gross weight of the shipping container shall be governed by the weight limitations of the applicable container specification.

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3.9.1.2 Type II. Components of kits and shop sets shall be packed in two or more exterior shipping containers. Items displacing three cubic feet or more shall be packed as specified in 3.9.1.2.1. Items displacing less than three cubic feet shall be packed as specified in 3.9.1.3. Compressed gas cylinders having a protective cap over the valve assembly may be shipped uncrated. Box closure and strapping requirements shall be in accordance with the applicable container specifications and the appendix thereto. In the event an open crate is utilized as the shipping container, a waterproof shroud shall be placed over the item. Anchoring, shrouding, blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186. (NOTE: Items packaged in accordance with method II requirements of MIL-P-116 shall not be packed in open crates).

3.9.1.2.1 Individual pack. All items displacing three cubic feet or more shall be individually packed. Shipping containers shall conform to PPP-B-585, class 3, styles 2 or 3; PPP-B-601, overseas type; PPP-B-621, class 2; MIL-C-52950; or MIL-C-3774. Consideration will be given the following factors when selecting a suitable shipping container:

a. Adequate protection for the commodity against physical damage.
b. The most economical container.
c. Minimum cube and weight.

3.9.1.3 Type III (consolidated pack). Packaged items displacing less than three cubic feet shall be consolidated and packed in a box conforming to PPP-B-636, class 2; PPP-B-601, overseas type; or MIL-B-43666, type I or type II. Selected items shall be placed in the shipping container in accordance with their fragility characteristics. Noncritical, rugged-type items shall be placed in the container first and critical, fragile-type items shall be positioned at the top of container. Blocking, bracing, and cushioning of the item shall be as specified in 3.9.1.2.

3.9.1.4 Consolidation container. Unless otherwise specified, compliance to individual packaging data sheets is waived when the tool set shipping containers conform to Style A of PPP-B-601 (overseas type), or class 2, style 2 or 4 of PPP-B-621 except the tops shall be 1/2" thick, plywood sheeting bonded with exterior type glue, or PPP-B-1055, waterproofed barrier material, covered with nominal 1" lumber. Sides and ends of PPP-B-621 boxes shall be lined with PPP-B-1055, waterproofed barrier material. The use of up to date packaging materials, such as cohesive VCI impregnated polyethylene, as an alternate to method IC, and cohesive coated cushioning materials, are encouraged.

3.9.1.5 Closure. Box closure, skids and strapping for shipping containers shall be in accordance with the appendix to the applicable box specification.

3.9.2 Level B. Types I, II, and III kits, tool sets, and shop sets shall be packed as specified for level A (see 3.9.1), except boxes utilized in shipping shall conform to PPP-B-601, domestic type, styles A, B, or J; PPP-B-621, class 1; or PPP-B-640, class 2 in lieu of those specified for level A. Closurers see (3.9.1.5).

3.9.3 Level C. Level C packing of tool sets, shop sets, and kits shall conform to the requirements of MIL-STD-1188.

3.9.4 Commercial packing. Commercial packing of tool sets, shop sets, and kits shall conform to the requirements of ASTM D 3951-82.

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3.9.5 Unitized load. When specified (see 6.2.1), a quantity of kits or sets shall be unitized in accordance with MIL-STD-147. A fiberboard cap shall be employed over the load as specified therein.

3.10 Performance.

3.10.1 Rough handling. When specified (see 6.2.1), the completed pack shall provide protection from rough handling sufficient to prevent damage to the packaged item(s) when the pack is subjected to the rough handling tests of MIL-P-116. Following the tests, the packaged item(s) shall show no evidence of damage, and the exterior container shall not reveal any failure caused by improper application or design of interior cushioning, blocking, or bracing. All barriers, wrappings, interior containers, blocking, bracing and anchoring shall be intact and capable of providing the intended protection.

3.10.2 Exposure. When specified (see 6.2.1) the completed pack shall provide protection from exposure to the elements sufficient to prevent damage to the packaged item(s) when the pack subjected to the cyclic exposure test of MIL-P-116. the test, the tested package, when opened, shall show no evidence of shall the packaged item show corrosion.

3.11 Marking.

3.11.1 Military markings. When military levels of protection are specified, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 and any special markings required by the contract or purchase order.

3.11.1.1 Cylinders. Painted surfaces of compressed gas cylinders shall not be defaced with markings except for the application of hazardous labels. Cylinders scheduled to be shipped uncrated shall be marked by the application of shipping tags. Identification markings and set markings shall be applied to tags conforming to UU-T-81, type A or B, class 2. Address markings shall be annotated on Military Shipping Tag DD Form 1387-1. To preserve the cylinders' identity and shipping address tags shall be attached to the protective cap of the cylinder with a metal wire tag. Wire shall be not less than 23 gage (0.0258-inch diameter). When cylinders are crated for shipment, identification markings shall be applied as specified above.

3.11.2 Packing list. When level A or C preservation is specified, an up to date packing list shall accompany each tool set, kit, or shop set. The packing list shall be prepared and utilized as specified in MIL-STD-129.

3.12 Storage. Items shall be stored in accordance with DOD 4145.19-R-1, Storage and Materials Handling.

3.13 Workmanship. Workmanship shall be such that when the proper procedure is followed, material and equipment being processed will receive the maximum protection against corrosion, deterioration, and damage during handling, shipment, and storage.

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4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements or purchase order as specified herein. Except as otherwise specified in the contract, or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such specifications are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet the requirements of section 3. The inspection set forth in this specification shall become a part of the contractors overall inspections systems or quality program. The absence of the responsibility of requirements in this specification shall not relieve the supplier of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance requirements, however, this does not authorize submission of known defective material, either indicted or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. When a first article sample is required by the contract or purchase order, the supplier shall subject the first article pack to the examinations in 4.3 and the applicable examinations and test in the unit pack design validation specified in the Quality Assurance Provisions of MIL-P-116 to determine conformance with the requirements of the specification. When specified (see 6.2.1), first articles packages shall be subjected to the rough handling and cyclic exposure test(s) (one or both) as contained in the unit pack design validation requirements of MIL-P-116. Failure of the first article to pass any of the tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied in all cases for acceptance under the contract or purchase order. Quality conformance inspection shall consist of the examinations in 4.3 and the applicable examinations and tests specified in the quality conformance inspection of MIL-P-116, with sampling to be conducted as specified in MIL-P-116. When specified (see 6.2.1), the package shall be subjected to rough handling and cyclic exposure test(s) (one or both) as contained in the quality conformance inspection requirements of MIL-P-116.

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4.3 Product examination. Visually, dimensionally, and manually examine each item to determine conformance with the requirements of 3.3 through 3.9.4 and 3.11 through 3.13. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Dimensional examination includes measuring dimensions as specified. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacturing at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operation. Any inspection of test performed in accordance with MIL-P-116 (see 4.2.1 and 4.2.2) does not need to be repeated. Failure of the contractor to provide objective evidence that the item and its components conform to all the preservation, packaging, packing, and marking standards and specifications cited in section 3 and have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

5 PACKAGING

5.1 This section is not applicable to this specification.

6 NOTES

6.1 Intended use. It is intended that this specification be used for reference in section 5 of related commodity specifications and direct reference in procurement documents. The preservation, packaging, and marking requirements specified herein are intended to ensure tools and equipment against corrosion, deterioration, and damage, and to provide for proper identification during shipment and storage.

6.2 Ordering data.

6.2.1 Acquisition Requirements. Acquisition documents should specify the following:

a. Title, number, and date of specification.
b. Type I, II, or III tools sets, kits, or shop sets, and equipment (see 1.2).
c. First article, when required (see 3.1).
d. Selection of applicable degree of protection A, B, C, or Commercial (see 3.7 and 3.9).
e. Unitized load, when required (see 3.9.5).
f. First article inspection, if different (see 4.2.1).
g. Quality conformance inspection, if different (see 4.2.2).
h. Rough handling test, when required (see 3.10.1 and 4.2.1).
i. Cyclic exposure test, when required (see 3.10.2 and 4.2.1).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts list and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.4 Tool sets and kits (small). Tool sets and kits furnished in plastic, cloth, or leather carrying cases, or relatively small wood or metal boxes, shall be packaged and packed in accordance with PPP-P-40.

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6.5 Unit protection. When the environment within tool set consolidation boxes (see 3.9.1.4) provides water-resistent protection to the items packed therein, the equivalent protection cited on individual packaging data sheets becomes excessive. Individual change request procedures to reduce packaging requirements are ineffective due to the universal magnitude. Certain items may still require Level A unit protection due to critical, sensitive or crushable characteristics.

6.6 First article. When a first article is required, it shall be tested and approved under the appropriate provision of the Federal Acquisition Regulation. The first article should be a preproduction sample, or a standard production item from the contractors current inventory. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, tests and approval of the first article.

6.7 Disposability methods. Environmental Pollution Preventative Measures are contained in the materials specifications indicated herein. Refer to the materials' specifications (or preparing activity) for recommended disposability methods.

6.8 Subject term (key word) listing.

1. Common tools
2. Packaging of
3. Hardware
4. Shipment

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AL
Navy - YD
Air Force - 99

Preparing activity:
Army - AL

Project No. PACK-0926

Review activities:
Army - AV, CR, SM
Air Force - 84

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-T-45542D, Tool Sets, Shop Sets and Kits Maintenance, Modification and

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER Tool Packaging of

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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